Semisupervised Semantic Segmentation of Remote Sensing Images With Foundation Model Distillation and Multilevel Consistency

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***Abstract*—Semantic segmentation of remote sensing images plays a crucial role in urban planning, environmental monitoring, and ecosystem management. However, its development is often constrained by the high cost of obtaining pixel-level annotations. Existing semisupervised learning (SSL) approaches mitigate this problem by leveraging a small amount of labeled data together with abundant unlabeled data. However, in scenarios with limited annotations, their performance is often degraded by pseudolabel noise and the insufficient utilization of structural information in unlabeled data. To address these challenges, we propose FDMatch, a semisupervised semantic segmentation framework for remote sensing images that integrates foundation model distillation with multilevel consistency learning. In particular, FDMatch employs a pretrained remote sensing foundation model (RSFM) as a knowledge source and transfers its rich spatial priors and high-level semantics to a lightweight segmentation network through an adaptive distillation mechanism, thereby enhancing the feature discrimination capability under limited supervision. Furthermore, a multilevel consistency regularization strategy is developed to overcome the limitation of conventional pixel-level consistency, which explores the potential structural information in unlabeled data by modeling interclass dependency patterns and interpixel correlations, thereby providing informative and reliable supervisory signals for model optimization. Comprehensive evaluations on three benchmark remote sensing datasets demonstrate that FDMatch consistently outperforms existing semisupervised methods across different label ratios, particularly exhibiting superior accuracy and robustness in scenarios with scarce labels.**

***Index Terms*—Consistency regularization, foundation model, knowledge distillation, remote sensing, semisupervised semantic segmentation.**

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I. Introduction

REMOTE sensing semantic segmentation aims to perform pixel-level classification to produce fine-grained geospatial representations, thereby supporting a wide range of applications such as urban monitoring, disaster assessment, and environmental mapping [1], [2], [3], [4]. In recent years, deep learning methods have become the dominant techniques in this field and have achieved remarkable progress. However, supervised approaches typically require large-scale, high-quality annotations, the acquisition of which is both labor-intensive and time-consuming [5], [6], [7]. In practical remote sensing applications, the scarcity of annotations caused by the complexity and heterogeneity of geospatial scenes makes improving model performance under limited supervision a major challenge. Semisupervised learning (SSL) mitigates this problem by integrating limited labeled samples with abundant unlabeled samples during training, thereby substantially reducing the reliance on manual annotations. Owing to its strong potential to enhance robustness and model generalization in large-scale remote sensing analysis with limited labeled samples, SSL has become a prominent research direction in semantic segmentation [8], [9], [10], [11].

Currently, mainstream semisupervised semantic segmentation methods can be categorized into self-training-based methods and consistency regularization-based methods [12]. Self-training-based methods typically train an initial model on a limited set of labeled samples, which is then used to generate pseudolabels for the unlabeled data. These pseudolabels are subsequently combined with the labeled set to iteratively refine the model's performance [13], [14], [15]. Consistency regularization-based methods enhance model robustness and representation learning by enforcing consistent predictions for the same input under different perturbations, such as data augmentations or noise injections [16], [17], [18], [19]. Although both methods have achieved promising results in remote sensing semantic segmentation, they still suffer from inherent limitations. These methods rely heavily on the quality of pseudolabels. In the early stages of training with limited labeled data, pseudolabels often contain substantial noise, leading to unstable optimization and degraded convergence. Recent methods, such as CorrMatch [20] and UCCL [21], alleviate this problem by exploiting pixel correlations or uncertainty-aware semantic cues. However, they still mainly mine supervision within the student/teacher prediction space.

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As a result, their performance remains sensitive to pseudolabel noise and is further constrained by backbones trained from scratch or pretrained on ImageNet, especially under scarce labeled data.

In recent years, remote sensing foundation models (RSFMs) have demonstrated remarkable feature extraction and generalization capabilities through large-scale pretraining on extensive remote sensing datasets [22], [23], [24], [25]. These models achieve competitive performance across various downstream tasks with only a small amount of labeled data for fine-tuning. Therefore, effectively leveraging the high-quality feature representations and predictive capability of foundation models becomes an important direction for improving semisupervised remote sensing semantic segmentation. However, foundation models typically involve a large number of parameters and substantial computational cost, which poses challenges to deployment and inference efficiency in resource-constrained real-world scenarios.

To address the aforementioned challenges, we propose FDMatch, a semisupervised semantic segmentation framework. Unlike conventional methods that rely on training from scratch or solely leverage ImageNet-pretrained backbones, FDMatch employs RSFMs as a source of general prior knowledge and introduces an adaptive foundation model distillation (AFMD) mechanism. By aligning the representation space of a lightweight segmentation network with the high-quality features of the foundation model, this mechanism enables the efficient transfer of knowledge from the foundation model to the lightweight segmentation network, thereby reducing the student model's reliance on noisy self-generated pseudolabels while maintaining low computational overhead. Moreover, to fully exploit the latent structural semantics in unlabeled data, we design a multilevel consistency regularization framework that constrains model predictions from two complementary perspectives: interclass relation consistency (ICRC) and interpixel correlation consistency (IPCC). Rather than treating classes independently, ICRC models the relative dependency or competition patterns among candidate classes in the predictive distribution and enforces their consistency across weak and strong augmentations. IPCC samples representative feature vectors, constructs similarity matrices, and encourages consistent pixel-wise relationships, thereby improving the model's ability to capture intrinsic spatial correlations. The main contributions of this work can be summarized as follows.

1) A semisupervised semantic segmentation framework for remote sensing images, FDMatch, is proposed to more effectively exploit unlabeled data by integrating AFMD with multilevel consistency regularization.
2) An AFMD mechanism is developed to dynamically align the feature representations of a lightweight segmentation network with those of an RSFM, providing stable guidance under sparse annotations.
3) A multilevel consistency strategy is designed to jointly enforce ICRC and IPCC, thereby improving semantic discrimination and spatial structure modeling.
4) Comprehensive experiments on three benchmark datasets demonstrate the effectiveness and generalization capability of the proposed framework across different labeled data ratios.

The remainder of this article is organized as follows. Section II reviews related studies on semisupervised semantic segmentation and RSFMs. Section III presents the proposed FDMatch framework in detail. Section IV describes the experimental datasets and comparative evaluation results. Section V provides further analysis, including ablation studies, training convergence, model generalization, advantages of the distillation strategy, and computational cost. Finally, Section VI concludes the study.

II. Related Works

A. Supervised Semantic Segmentation

Semantic segmentation assigns a category to each pixel in an image and has achieved substantial progress with the advancement of deep learning. Early convolution-based methods such as FCN [26], U-Net [27], and DeepLabv3+ [28] advanced end-to-end learning, encoder–decoder fusion, and multiscale context modeling. However, the local receptive fields of CNNs limit their ability to model long-range dependencies. Transformer-based architectures address this issue by capturing global context, with SegFormer [29] integrating a Transformer encoder and a lightweight MLP decoder for efficient multiscale fusion.

However, the performance of both CNN- and Transformer-based methods depends heavily on large-scale, high-quality annotations, which are costly to obtain in pixel-level segmentation and often unavailable in practice. Semisupervised semantic segmentation alleviates this limitation by combining limited labeled samples with abundant unlabeled data, significantly reducing annotation cost while preserving competitive performance and thus has become an increasingly promising research direction.

B. Semisupervised Semantic Segmentation

The core idea of SSL is to exploit potential supervisory signals from unlabeled data based on a limited number of labeled data, thereby reducing the model's reliance on annotated data. Existing mainstream approaches can be categorized into two groups: self-training-based methods and consistency regularization-based methods. Self-training-based methods typically begin by training an initial model on labeled data and then generating pseudolabels for unlabeled data. High-confidence predictions are selected to participate in subsequent retraining, thereby improving the model's performance [15], [30], [31], [32].

Consistency regularization-based methods enhance model robustness by enforcing consistent predictions under different perturbations of the same input image. Cross consistency training (CCT) [18] enhances stability by constraining consistency between a primary decoder and multiple auxiliary decoders under feature perturbations. To further improve the utilization of unlabeled data, several approaches integrate weak-to-strong consistency into the self-training framework. FixMatch [33] uses weakly augmented predictions as pseudolabels to supervise strongly augmented views, while UniMatch [34] further introduces feature-level perturbations to strengthen consistency learning. However, these methods primarily operate at the pixel level and struggle to capture structured semantics. To address this issue, recent studies extend consistency modeling

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to interpixel and semantic-level relations. RankMatch [35] enforces rank-based correlation consistency via proxy points, CorrMatch [20] propagates labels through pixel-pair similarity to refine pseudolabels, and UCCL [21] extracts consistent semantic cues from uncertain regions to achieve class-level alignment. In the field of remote sensing imagery, DWL [9] reduces pseudolabel noise through a decoupled weighted learning framework and a ranking-based mechanism to mitigate class imbalance.

Although these methods improve semisupervised segmentation from different perspectives, they still mainly exploit supervision generated within the current model prediction space. Consequently, their performance remains highly dependent on the representation quality of the backbone and is still vulnerable to noisy pseudolabels in low-label remote sensing scenarios. To address these issues, this study incorporates an RSFM as prior knowledge and introduces a collaborative framework that integrates knowledge distillation with multilevel consistency constraints, thereby enhancing pseudolabel reliability and stabilizing the training process.

C. Remote Sensing Vision Foundation Model

Foundation models are large-scale pretrained models that provide transferable feature representations for diverse downstream tasks [36], [37], [38]. Trained with advanced paradigms such as self-supervised learning, they learn rich feature patterns from massive data and exhibit strong cross-domain transferability in NLP and computer vision [39], [40], [41], [42]. However, remote sensing imagery differs markedly from natural images in imaging geometry, spatial–temporal resolution, and spectral characteristics, causing models pretrained on natural images to generalize poorly [43]. To bridge this gap, remote sensing-specific foundation models have been developed by incorporating temporal, multispectral, and multitask information [44], [45]. Among them, MTP [46] adopts a unified pretraining framework with a shared encoder and multiple task-oriented decoders, enabling more general and transferable remote sensing representations.

Despite their strong representational capacity, foundation models are often computationally heavy, limiting their direct use in semisupervised segmentation. To address this, we adopt a knowledge distillation strategy to transfer the rich remote sensing priors encoded in foundation models into a lightweight segmentation network, enhancing performance without increasing inference cost.

III. Methodology

In this section, we provide a detailed description of the proposed FDMatch framework. First, an AFMD strategy is introduced to transfer the general knowledge embedded in the pretrained RSFM to a lightweight segmentation network, thereby alleviating the performance bottleneck caused by limited labeled data. Subsequently, a multilevel consistency constraint is constructed from two perspectives— ICRC and IPCC—to fully exploit the latent structural semantic information within unlabeled data and enhance the model's representation capability for complex remote sensing scenes. The overall architecture of FDMatch is illustrated in Fig. 1.

A. Preliminaries

FDMatch is built upon the classical strong–weak consistency regularization framework [34]. Given a labeled dataset $D_l = \{(x_i^l, y_i^l)\}_{i=1}^{N^l}$ and an unlabeled dataset $D_u = \{x_i^u\}_{i=1}^{N^u}$, the goal of semisupervised semantic segmentation is to train a segmentation model $\mathcal{F}$ by jointly leveraging a limited number of labeled data and a large amount of unlabeled data, where N_l and N_u denote the numbers of labeled and unlabeled images, respectively, with $N_l \ll N_u$. The framework adopts a teacher–student architecture, in which the teacher network either shares parameters with the student network or serves as an exponential moving average (EMA) version of it. For the labeled data, supervised learning is performed by minimizing the cross-entropy loss, which is defined as follows:

$$\mathcal{L}_{\text{sup}} = \frac{1}{N_l}\sum_{i=1}^{N_l}\frac{1}{HW}\sum_{j=1}^{HW}\ell_{ce}\left(\mathcal{F}\left(x_{ij}^l\right), y_{ij}^l\right) \tag{1}$$

where H and W denote the height and width of the input image, respectively; ℓ_{ce} represents the standard pixel-wise cross-entropy loss. For the unlabeled data, weakly augmented versions x_i^{wu} and strongly augmented versions x_i^{su} are generated and fed into the model to obtain the corresponding prediction results (p^{wu} and p^{su}), where the model outputs from the weakly augmented images serve as pseudolabels for computing the unsupervised loss

$$\mathcal{L}_u = \frac{1}{N_u}\sum_{i=1}^{N_u}\frac{1}{HW}\sum_{j=1}^{HW}\ell_{ce}\left(p_{ij}^{su}, p_{ij}^{wu}\right). \tag{2}$$

Building upon this baseline framework, this study proposes AFMD and multilevel consistency regularization to further enhance the performance and stability of SSL in remote sensing image segmentation.

B. Adaptive Foundation Model Distillation

RSFMs, pretrained on large-scale datasets, are capable of learning generalized and transferable visual representations. With only a small number of labeled samples for fine-tuning, they can achieve remarkable performance on downstream tasks. To fully exploit the representational knowledge of foundation models and effectively transfer it to lightweight segmentation networks commonly used in SSL, we propose an AFMD method. The proposed approach consists of two stages: the knowledge adaptation stage and the knowledge distillation stage.

1) Knowledge Adaptation Stage: Due to the inherent differences in network architecture and parameter scale between the foundation model and the lightweight segmentation network, the two models learn feature representations in different spaces. This representation gap makes direct knowledge transfer from the foundation model to the lightweight network less effective. Therefore, this stage aims to bridge the feature-space discrepancy between the two models before distillation.

In particular, the lightweight segmentation network is first pretrained using limited labeled data to obtain its task-specific representation space. The segmentation head of the foundation model is then replaced with a segmentation head that shares the same structure and parameters as the pretrained lightweight network. Meanwhile, a feature projection layer is introduced

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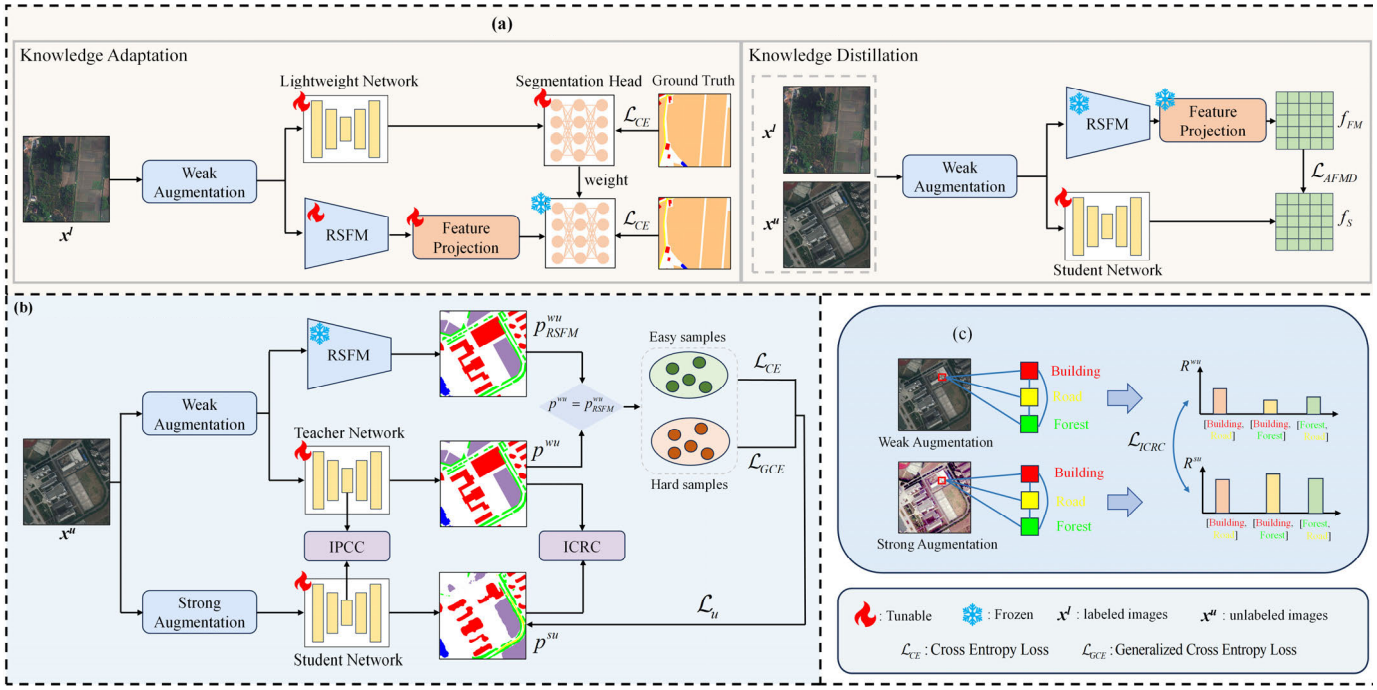


Fig. 1. Framework of the FDMatch. (a) AFMD transfers the knowledge from a pretrained RSFM to a lightweight segmentation network within the SSL framework. (b) Multilevel consistency regularization extends conventional pixel-level consistency by introducing ICRC and IPCC. (c) ICRC constructs consistency constraints by modeling the relative dependencies and competitive relationships among candidate classes within the predictive distribution.

to map foundation model features into the feature space of the lightweight network. Although the foundation model has strong generalization capability from large-scale remote sensing pretraining, its representations are not fully aligned with the target segmentation task. Therefore, the foundation model is further fine-tuned using limited labeled data to adapt it to the downstream task and reduce the representation discrepancy between the two models. This adaptation establishes an effective cross-model feature mapping relationship and provides more task-relevant knowledge for subsequent distillation. After this adaptation stage, the foundation model is frozen and used as a fixed knowledge source during FDMatch training.

2) Knowledge Distillation Stage: During semisupervised training, a performance-aware distillation strategy is designed to achieve dynamic knowledge transfer from the foundation model to the student network. To ensure training stability, the parameters of the foundation model are kept fixed, and its segmentation head, which is consistent with that used in the knowledge adaptation stage, is also frozen throughout the training process. At the feature level, both labeled and unlabeled data are fed into the foundation model and the student network to obtain their respective feature representations, denoted as f_{FM} and f_S. These features are then projected into a shared embedding space, where feature alignment is enforced using the MSE loss

$$\mathcal{L}_{AFMD} = \|f_{FM} - f_S\|^2. \tag{3}$$

Considering that the foundation model is fine-tuned using only a limited amount of labeled data, its performance may gradually lag behind the student network during training. To avoid potential negative knowledge transfer, a performance-aware knowledge transfer strategy is designed to selectively activate feature alignment. For labeled data, feature alignment is enforced only when the cross-entropy loss of the foundation model is lower than that of the student network. For unlabeled data, prediction entropy is adopted as a confidence measure [47], and feature alignment is performed when the entropy of the foundation model is lower than that of the student network. This design ensures that knowledge transfer from the foundation model to the student network occurs only when the former provides more reliable supervision, thereby improving the stability and effectiveness of the distillation process.

Moreover, considering the varying learning difficulty of samples and the potential noise in pseudolabels, unlabeled samples are divided into two groups according to the consistency between the predictions of the foundation model p^{wu}_{RSFM} and those of the teacher model p^{wu}. Samples with consistent predictions are regarded as easy samples and optimized using the standard cross-entropy loss, while samples with inconsistent predictions are regarded as hard samples, which are more likely to contain noisy pseudolabels and are therefore modeled using the more robust generalized cross-entropy loss [48]. Accordingly, the unsupervised loss $\mathcal{L}_u$ is redefined as follows:

$$\mathcal{L}_u = \frac{1}{N_u}\sum_{i=1}^{N_u}\frac{1}{HW}\sum_{j=1}^{HW}\left(\alpha_{ij}\ell_{ce}\left(p^{su}_{ij}, p^{wu}_{ij}\right) + \left(1-\alpha_{ij}\right)\ell_{gce}\left(p^{su}_{ij}, p^{wu}_{ij}\right)\right) \tag{4}$$

where $\alpha_{ij} \in \{0, 1\}$ denotes the sample difficulty weight (one for consistent predictions and zero otherwise). This strategy balances supervision across samples of varying difficulty and improves training stability under label noise.

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C. Interclass Relation Consistency

Consistency regularization based on weak–strong augmentations typically uses predictions from the weakly augmented view as pseudolabels to supervise the strongly augmented view, thereby enforcing output consistency. However, most existing methods rely solely on the most confident class and treat classes independently, overlooking the relational structure among candidate classes in the predictive distribution. For instance, low vegetation and trees are often more easily confused than trees and buildings, indicating stronger competition between similar classes in model predictions. Ignoring such interclass dependencies limits the effective exploitation of structured information in unlabeled data. To address this limitation, we introduce an ICRC constraint that maintains both prediction consistency and the stability of interclass dependency patterns across augmentations.

In particular, we construct a soft metric based on information entropy to characterize interclass dependencies. Rather than directly measuring intrinsic semantic relationships, this metric serves as a soft proxy for the relative dependency or competition among candidate classes in the predictive distribution. When the model encounters more distinct class pairs, it typically produces confident predictions with low uncertainty. On the contrary, for similar classes, their overlap in feature space makes them harder to distinguish, causing the predictive probability to be distributed across multiple classes and resulting in relatively higher entropy. For instance, given pixels belonging to trees, the model is more likely to confuse them with similar categories such as low vegetation than with distant categories such as cars. This leads to higher probability mass assigned to similar classes and consequently higher entropy. We note that high entropy does not necessarily indicate true semantic relatedness, since it may also arise from insufficient learning or noisy predictions, especially in early training stages. Therefore, in our method, entropy is not interpreted as an exact semantic quantity; instead, we focus on the relative dependency structure induced by the predictive distribution. This structure is generally more stable across different augmented views of the same sample than the exact entropy values themselves.

We first select the top-K (empirically, K = 4) classes with the highest predicted probabilities from the weakly augmented view and then compute the information entropy $R(u, v)$ between each class pair (u, v) using the prediction results from both weak and strong augmentations

$$R(u,v) = -\left(\frac{p_u}{p_u+p_v}\log\frac{p_u}{p_u+p_v} + \frac{p_v}{p_u+p_v}\log\frac{p_v}{p_u+p_v}\right) \tag{5}$$

where p_u denotes the predicted probability of class u. Because unlabeled samples lack ground truth, enforcing strict numerical alignment of interclass relationships may introduce noise. Therefore, we adopt a soft-ranking consistency strategy, which preserves only the relative ordering of interclass relationship values across augmentations, improving robustness. The soft-ranking score of the kth class pair (u_k, v_k) is denoted as $r(k)$

$$r(k) = \sum_{l=1}^{C_K^2} \sigma\left(1 + R(u_l, v_l) - R(u_k, v_k)\right) \tag{6}$$

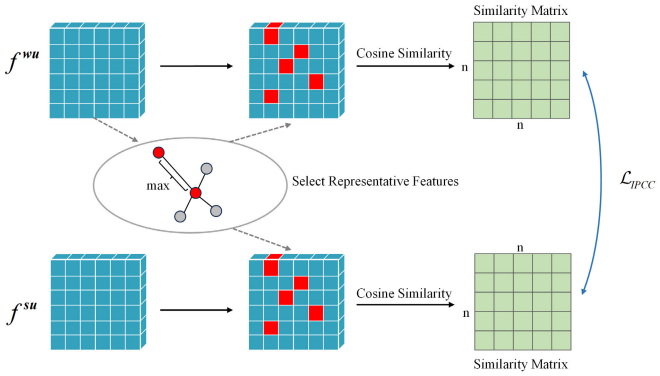


Fig. 2. Illustration of the IPCC, where the consistency constraint is established by modeling the correlations among pixels.

where σ denotes the sigmoid function. The ranking loss is then computed as follows:

$$\mathcal{L}_{rank} = \frac{1}{N_u}\sum_{i=1}^{N_u}\frac{1}{K}\sum_{k=1}^{C_K^2}\frac{1}{HW}\sum_{j}^{HW}\left(1 - \frac{\min\left\{r_{ij}^{w}(k), r_{ij}^{s}(k)\right\}}{\max\left\{r_{ij}^{w}(k), r_{ij}^{s}(k)\right\}}\right). \tag{7}$$

Moreover, since a higher dependency value indicates that the model tends to assign probability mass to multiple competing classes, we expect the model to produce more discriminative predictions. To this end, an auxiliary loss is proposed to minimize the dependency between the most confident class and the others, thereby encouraging the model to enhance class separability while preserving interclass dependency modeling. The loss function is defined as follows:

$$\mathcal{L}_{aux} = \frac{1}{N_u}\sum_{i=1}^{N_u}\frac{1}{HW}\sum_{j}^{HW}\sum_{c\neq y^*}^{C} R\left(y_{ij}^{*}, c\right) \tag{8}$$

where y^* denotes the class with the highest predicted probability and C represents the total number of classes. Finally, the ICRC loss can be formulated as follows:

$$\mathcal{L}_{ICRC} = \mathcal{L}_{rank} + \mathcal{L}_{aux}. \tag{9}$$

D. Interpixel Correlation Consistency

To further leverage the supervisory signals in unlabeled data, we extend consistency regularization from pixel-level predictions to pixel-wise correlation modeling. The core idea is to select a set of representative feature vectors, compute their pairwise similarities to form a correlation matrix, and enforce its consistency across weak and strong augmentations, thereby preserving the relational structure among pixels, as illustrated in Fig. 2.

In the feature selection stage, a set of representative and diverse feature vectors is first extracted from the high-dimensional feature space. In particular, given the feature map of the weakly augmented image $f^{wu} \in \mathbb{R}^{H\times W\times C}$, the mean vector of all feature vectors is computed as the feature center. Subsequently, based on cosine similarity, the feature vector that is least similar to the center is selected as the first representative vector v_1, to avoid proximity to the global mean feature and thus ensure information diversity. When selecting

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TABLE I

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE WHDLD DATASET. OPTIMAL RESULTS ARE INDICATED IN BOLD. ALL METRICS ($F1$, MEANF1, MIOU) ARE REPORTED AS PERCENTAGES (%)

Ratio	Method	F1						MeanF1	mIoU
		building	road	pavement	vegetation	bare soil	water		
1% (34:3424)	Onlysup	61.05	51.42	49.67	85.53	43.04	92.45	63.86	49.95
	Onlysup MTP	64.84	62.52	47.35	85.54	48.11	92.62	66.83	52.85
	FixMatch [33]	65.79	59.71	40.78	85.71	47.70	94.62	65.72	52.22
	UniMatch [34]	58.80	61.17	50.50	84.93	49.71	94.82	66.66	52.75
	CorrMatch [20]	63.74	60.97	45.52	85.70	46.66	94.75	66.82	53.25
	RankMatch [35]	65.71	62.07	41.05	83.56	49.73	91.71	65.64	51.55
	DWL [9]	62.84	61.64	52.76	86.17	47.86	94.63	67.65	53.86
	UCCL [21]	68.46	62.91	47.60	85.20	48.74	92.12	67.50	53.50
	FDMatch	**68.92**	**66.76**	**54.45**	**87.68**	**51.40**	**94.97**	**70.70**	**57.19**
5% (172:3286)	Onlysup	69.27	63.30	52.69	87.43	49.07	94.84	69.43	55.90
	Onlysup MTP	70.32	68.09	54.77	87.62	53.03	95.17	71.50	58.06
	FixMatch [33]	71.13	68.14	54.82	87.78	47.98	96.12	71.00	57.83
	UniMatch [34]	71.09	67.75	54.24	87.59	52.40	96.09	71.53	58.25
	CorrMatch [20]	70.50	68.59	53.10	88.00	46.99	96.17	70.56	57.45
	RankMatch [35]	71.79	67.44	55.33	88.04	53.16	96.23	72.00	58.78
	DWL [9]	71.17	68.51	55.72	88.61	51.59	96.42	72.00	58.89
	UCCL [21]	72.03	68.75	53.68	88.16	49.59	96.19	71.40	58.30
	FDMatch	**72.51**	**69.73**	**56.60**	**88.68**	**54.20**	**96.51**	**73.04**	**59.99**
10% (345:3113)	Onlysup	70.84	68.05	56.12	88.46	49.21	95.86	71.42	58.24
	Onlysup MTP	71.69	69.93	56.91	88.12	51.54	95.78	72.33	59.13
	FixMatch [33]	72.01	69.36	56.06	88.65	51.50	96.51	72.35	59.31
	UniMatch [34]	72.48	70.44	56.97	88.53	50.40	96.23	72.51	59.48
	CorrMatch [20]	71.25	68.62	54.78	88.21	48.19	96.16	71.20	58.09
	RankMatch [35]	73.08	70.94	56.84	88.86	52.01	**96.72**	73.07	60.17
	DWL [9]	71.49	70.48	58.96	88.82	52.62	96.38	73.12	60.07
	UCCL [21]	**73.24**	70.05	57.66	88.70	51.74	96.66	73.01	60.05
	FDMatch	73.09	**71.04**	**58.50**	**88.90**	**53.33**	96.54	**73.56**	**60.62**

the kth representative vector, the existing representative set is denoted as $V_{k-1} = \{v_1, v_2, \ldots, v_{k-1}\}$. For each unselected feature vector f_j, its maximum similarity s_{f_j} with all vectors in V_{k-1} is computed, and the feature vector with the smallest s_{f_j} value is selected as the kth representative vector v_k, thereby maximizing the diversity of the selected features

$$s_{f_j} = \max_{v \in V_{k-1}} \frac{f_j \cdot v}{\|f_j\| \, \|v\|}$$
$$v_k = \arg \min_{f_j \notin V_{k-1}} s_{f_j}. \tag{10}$$

Repeat the above process until n representative vectors are selected. Based on these representative vectors, we extract feature vectors from the corresponding positions of the weakly augmented feature map f^{wu} and the strongly augmented feature map f^{su}, respectively. Then, we construct the similarity matrices $C^w, C^s \in \mathbb{R}^{n \times n}$, where each element C_{ij} denotes the cosine similarity between the corresponding representative vectors.

Finally, the similarity consistency between the weakly and strongly augmented views is enforced by minimizing the KL divergence between their similarity matrices. The loss function is defined as follows:

$$\mathcal{L}_{IPCC} = \frac{1}{N_u} \sum_{i=1}^{N_u} \frac{1}{HW} \sum_{j}^{HW} \ell_{kl}\left(C_{ij}^s, C_{ij}^w\right). \tag{11}$$

This constraint preserves interpixel correlation structure and improves feature discriminability and robustness. The final training objective integrates all components

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \mathcal{L}_u + \lambda_1 \mathcal{L}_{AFMD} + \lambda_2 \mathcal{L}_{ICRC} + \lambda_3 \mathcal{L}_{IPCC} \tag{12}$$

where λ_1, λ_2, and λ_3 are weighting coefficients balancing the loss terms, all empirically set to 0.1 in our experiments.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets and Evaluation Metrics

To evaluate the effectiveness of the proposed FDMatch framework in semisupervised semantic segmentation tasks, we conduct experiments on three representative remote sensing datasets: the WHDLD [49], LoveDA [50], and ISPRS Vaihingen.

1) WHDLD: This dataset contains 4940 RGB images with a spatial resolution of 2 m and a size of 256 × 256 pixels. It is randomly split into 3458 training, 494 validation, and 988 testing samples following a 7:1:2 ratio.

2) LoveDA: This large-scale land-cover dataset comprises 5987 images from three cities, including 166 768 annotated objects. Each 1024 × 1024 image (0.3 m resolution) is cropped into nonoverlapping 512 × 512 patches, producing 17 564 samples. These are divided into 10 088 training, 3338 validation, and 3338 testing samples according to a 6:2:2 split.

3) ISPRS Vaihingen: The dataset consists of 33 orthophotos (1996 × 1995 to 3816 × 2550 pixels, 9 cm resolution), of which 15, 1, and 17 are used for training, validation, and testing. All images are cropped into 512 × 512 patches with a 256-pixel overlap to facilitate model training.

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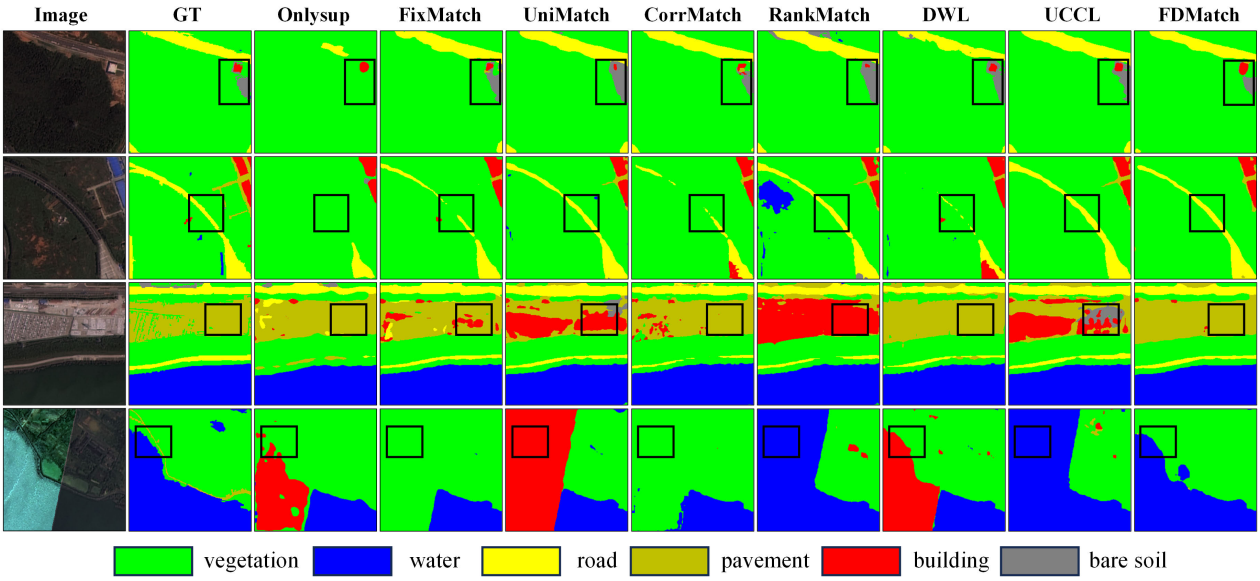


Fig. 3. Visual results of the proposed method are compared with those of the only-supervised baseline and other advanced semisupervised methods on the 1% labeled WHDLD dataset.

To evaluate the performance of the proposed method, the $F1$ score and the mean intersection over union (mIoU) are adopted as evaluation metrics.

B. Experimental Setting

All experiments are implemented on two NVIDIA RTX 4090 GPUs using the PyTorch framework. We adopt DeepLabv3+ with a ResNet-50 backbone as the basic segmentation architecture and employ MTP [46] as the RSFM for knowledge transfer. The optimizer is set to stochastic gradient descent (SGD) with an initial learning rate of 0.01, a weight decay of 0.0001, and a momentum of 0.9. The model is trained for a maximum of 80 epochs, and the learning rate is gradually decayed following the poly scheduling strategy. For data preprocessing, we follow the same augmentation strategy as UniMatch, including random cropping, color jittering, and horizontal flipping. To evaluate the proposed method under different label rates, we construct training environments with 1%, 5%, and 10% labeled data on each dataset, using the corresponding labeled data and the remaining unlabeled data for semisupervised training.

C. Results and Comparison With State-of-the-Art Methods

This section systematically evaluates the performance of FDMatch on three remote sensing datasets and provides a comprehensive comparison with several existing semisupervised semantic segmentation methods. The compared algorithms include FixMatch [33], UniMatch [34], CorrMatch [20], RankMatch [35], DWL [9], and UCCL [21], all reproduced using their officially released implementations for a fair comparison. In addition, we establish an Onlysup baseline, where the segmentation network is trained solely with labeled data under supervised learning, to quantify the performance gain achieved by semisupervised algorithms. Meanwhile, we further introduce a stronger baseline, Onlysup MTP, which directly fine-tunes the foundation model on labeled samples.

1) WHDLD Dataset: Table I reports the results on the WHDLD dataset under different labeled ratios. FDMatch consistently attains the highest MeanF1 and mIoU across all settings. Under the 1%, 5%, and 10% labeling ratios, it exceeds the Onlysup baseline by 7.24%, 4.09%, and 2.38% mIoU, respectively, showing particularly large gains when annotations are extremely limited. Compared with the second-best semisupervised method, FDMatch improves mIoU by 3.33%, 1.10%, and 0.45%, with the advantage increasing as the labeled ratio decreases. Although DWL alleviates pseudolabel noise through decoupled supervision and thus achieves the second-best performance, it remains restricted by insufficient labeled signals. Notably, the strong baseline Onlysup MTP, which directly fine-tunes the foundation model using only labeled data, outperforms several semisupervised methods that leverage unlabeled data, indicating the strong representation capability of RSFMs. On the contrary, FDMatch effectively strengthens feature representation by distilling generalized priors from the RSFM, demonstrating the benefit of foundation model knowledge transfer in semisupervised segmentation. Qualitative results under the 1% labeled ratio further support this conclusion, as shown in Fig. 3. Competing methods fail to accurately delineate small bare-soil regions and often produce fragmented predictions for narrow roads, whereas FDMatch yields clearer boundaries and more continuous structures. Furthermore, FDMatch maintains reliable predictions for categories that are susceptible to misclassification caused by spectral variations, demonstrating strong robustness in complex land-cover segmentation scenarios.

2) LoveDA Dataset: Table II reports the results on the LoveDA dataset under different labeled ratios. FDMatch consistently achieves the highest MeanF1 and mIoU across all settings. As a highly challenging dataset characterized by multiscale objects, complex background interference, and significant urban–rural domain shifts, LoveDA imposes strong requirements on model generalization. Under the 1%, 5%, and 10% labeled ratios, FDMatch surpasses the Onlysup baseline

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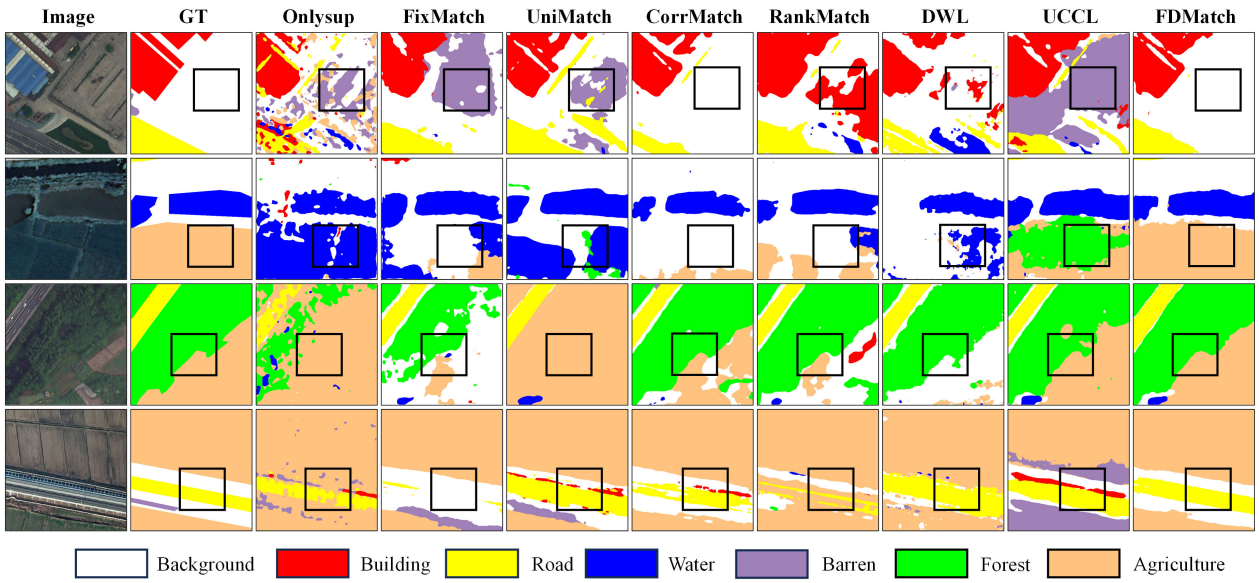


Fig. 4. Visual results of the proposed method are compared with those of the only-supervised baseline and other advanced semisupervised methods on the 1% labeled LoveDA dataset.

TABLE II

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE LOVEDA DATASET. OPTIMAL RESULTS ARE INDICATED IN BOLD. ALL METRICS ($F1$, MEANF1, MIOU) ARE REPORTED AS PERCENTAGES (%)

Ratio	Method	F1							MeanF1	mIoU
		Background	Building	Road	Water	Barren	Forest	Agriculture		
1% (100:9988)	Onlysup	53.80	56.72	40.79	47.85	27.34	40.15	58.44	46.44	30.81
	Onlysup MTP	54.07	69.57	70.13	57.88	42.89	51.12	65.86	58.93	42.40
	FixMatch [33]	61.20	62.07	60.86	73.99	34.77	36.32	63.35	56.08	40.16
	UniMatch [34]	52.59	67.82	**66.39**	78.78	37.89	25.60	64.65	56.25	41.07
	CorrMatch [20]	**63.69**	63.80	63.46	**80.05**	27.76	46.43	67.70	58.98	43.47
	RankMatch [35]	55.67	68.39	63.84	79.57	37.88	43.08	68.94	59.62	43.86
	DWL [9]	58.74	63.64	64.11	71.70	41.13	43.33	63.60	58.03	41.64
	UCCL [21]	53.48	65.71	64.75	75.97	35.25	33.04	64.25	56.06	40.44
	FDMatch	62.86	**72.53**	66.27	78.88	**46.50**	**54.87**	**70.48**	**64.63**	**48.56**
5% (504:9584)	Onlysup	61.97	61.78	48.06	60.92	25.48	49.75	61.45	52.77	36.73
	Onlysup MTP	63.61	72.85	**70.57**	75.14	41.42	55.95	68.20	63.96	47.91
	FixMatch [33]	68.14	70.34	65.81	76.97	38.25	50.86	68.60	62.71	46.79
	UniMatch [34]	67.27	74.51	68.03	79.03	33.69	55.58	66.88	63.57	47.99
	CorrMatch [20]	68.22	73.26	67.00	77.93	40.89	55.20	67.73	64.19	48.24
	RankMatch [35]	66.86	**74.95**	68.05	78.00	34.29	55.83	65.64	63.38	47.71
	DWL [9]	66.99	69.53	64.84	76.74	38.42	56.31	62.65	62.21	46.07
	UCCL [21]	66.50	74.30	69.65	76.36	24.90	49.34	**69.07**	61.45	46.27
	FDMatch	**68.29**	73.37	67.97	**81.91**	**43.43**	**58.18**	67.13	**65.75**	**49.99**
10% (1008:9080)	Onlysup	62.56	65.98	53.14	61.25	40.59	48.56	61.13	56.17	39.52
	Onlysup MTP	66.42	75.82	**71.01**	77.77	43.06	56.80	68.12	65.57	49.74
	FixMatch [33]	64.84	70.32	70.33	81.57	37.93	54.31	**71.60**	64.41	48.82
	UniMatch [34]	68.90	**76.81**	69.74	80.38	45.80	55.66	69.04	66.62	50.95
	CorrMatch [20]	69.38	68.71	69.00	**82.22**	46.88	53.55	69.80	65.65	49.82
	RankMatch [35]	68.24	76.35	68.43	80.62	42.93	**56.47**	70.28	66.19	50.56
	DWL [9]	66.65	70.63	68.00	76.34	46.15	52.49	63.30	63.36	47.10
	UCCL [21]	66.18	73.84	70.14	77.67	37.17	52.44	68.85	63.76	48.05
	FDMatch	**69.41**	74.89	70.61	80.55	**49.03**	52.67	70.29	**66.78**	**51.06**

by 18.19%, 12.98%, and 10.61% in MeanF1, demonstrating its ability to effectively leverage latent supervision from unlabeled data. Compared with the second-best semisupervised methods, FDMatch further improves MeanF1 by 5.01%, 1.56%, and 0.16%, with more pronounced gains under scarce-label scenarios. From a class-level perspective, FDMatch also shows clear advantages on minority categories. Under the 1% labeling setting, the $F1$ scores of barren and forest increase by 5.37% and 8.44%, indicating strong robustness and discriminative ability even under severe class imbalance. Fig. 4 provides qualitative comparisons under the 1% labeled ratio. On this multiclass and highly heterogeneous dataset, FDMatch

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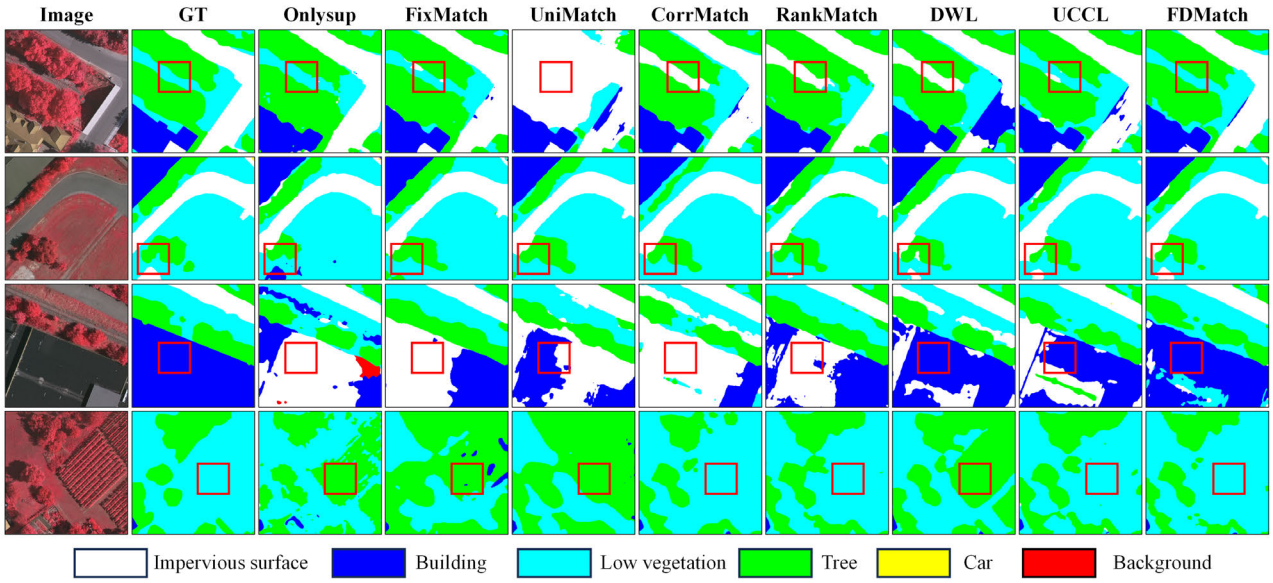


Fig. 5. Visual results of the proposed method are compared with those of the only-supervised baseline and other advanced semisupervised methods on the 1% labeled Vaihingen dataset.

produces segmentation maps with higher structural integrity and less noise, significantly outperforming other methods. These results further validate the effectiveness of the proposed approach in complex and diverse remote sensing scenarios.

3) ISPRS Vaihingen Dataset: Table III reports the results on the Vaihingen dataset under different labeled ratios. The clutter/background class is excluded from evaluation due to its extremely small proportion and ambiguous boundaries, which provide limited value for performance assessment. Across the 1%, 5%, and 10% labeling settings, FDMatch consistently achieves the highest MeanF1 and mIoU, outperforming both SOTA semisupervised approaches and the Onlysup baseline. Under the challenging 1% setting, FixMatch suffers from severe pseudolabel noise and performs poorly among comparison algorithms because image-level perturbation consistency alone cannot provide stable supervision with very limited labeled samples. UniMatch alleviates this issue through feature-level perturbations, while RankMatch, CorrMatch, and UCCL enhance pseudolabel reliability by exploiting pixel similarity or prediction uncertainty. The proposed FDMatch simultaneously models pixel-wise correlations and interclass relational consistency, enabling more effective exploitation of structured semantics in unlabeled data and thereby achieving superior performance across all ratios. Fig. 5 presents the qualitative comparisons of different methods. FDMatch achieves higher prediction accuracy in distinguishing categories with similar spectral characteristics and high interclass confusion, such as low vegetation and trees, as well as buildings and impervious surfaces. These results demonstrate the strong semantic discrimination capability of FDMatch in complex remote sensing scenarios.

V. Discussion

A. Ablation Experiments

1) Effects of Each Module of FDMatch: The proposed FDMatch framework consists of three core components:

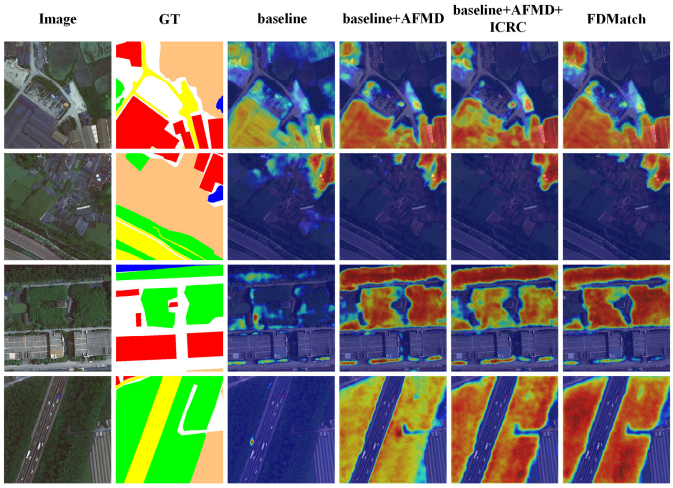


Fig. 6. Class activation maps of the building (first and second rows) and forest (third and fourth rows) categories under different component ablation settings.

AFMD, ICRC, and IPCC. All these components are built upon and extended from the strong–weak consistency framework of UniMatch. To evaluate the effectiveness of each component, we conducted ablation studies on the WHDLD and LoveDA datasets under the 1% labeled ratio, with the results summarized in Table IV. In particular, incorporating AFMD leads to mIoU improvements of 2.53% on WHDLD and 4.08% on LoveDA, validating that distilling knowledge from the foundation model significantly enhances performance in limited-label scenarios. Furthermore, ICRC and IPCC extend the conventional strong–weak consistency regularization by explicitly modeling semantic structural information from interclass relations and interpixel correlations. These modules improve the mIoU by 1.93% and 1.34% on WHDLD, and by 1.95% and 2.23% on LoveDA.

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TABLE III

PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE ISPRS VAIHINGEN DATASET. OPTIMAL RESULTS ARE INDICATED IN BOLD. ALL METRICS ($F1$, MEANF1, MIOU) ARE REPORTED AS PERCENTAGES (%)

Ratio	Method	F1					MeanF1	mIoU
		Imp.suf	Building	Lowveg	Tree	Car		
1% (9:967)	Onlysup	80.21	83.15	67.61	79.35	53.81	72.83	58.35
	Onlysup MTP	84.15	87.54	67.80	80.49	31.82	70.36	57.61
	FixMatch [33]	83.07	87.03	65.15	80.28	60.21	75.15	61.30
	UniMatch [34]	84.77	90.19	68.95	79.75	68.40	78.41	65.32
	CorrMatch [20]	86.57	90.84	74.62	83.88	62.22	79.63	67.29
	RankMatch [35]	86.26	90.59	72.72	82.52	65.17	79.45	66.87
	DWL [9]	**86.99**	**91.59**	**76.23**	83.84	60.05	79.74	67.62
	UCCL [21]	85.86	91.26	73.53	82.21	**70.80**	80.73	68.38
	FDMatch	86.97	90.97	75.59	**83.84**	68.20	**81.11**	**69.01**
5% (48:928)	Onlysup	85.43	89.02	75.88	84.54	65.67	80.11	67.60
	Onlysup MTP	87.69	90.69	76.70	84.23	66.21	81.10	69.10
	FixMatch [33]	87.78	91.66	77.46	85.06	73.43	83.08	71.61
	UniMatch [34]	88.30	93.08	78.84	84.67	76.29	84.24	73.26
	CorrMatch [20]	88.21	92.65	78.44	85.59	70.53	83.09	71.81
	RankMatch [35]	88.98	93.34	78.71	85.44	75.47	84.39	73.55
	DWL [9]	89.00	93.33	78.46	85.41	**77.43**	84.72	73.99
	UCCL [21]	88.71	93.01	78.08	85.32	75.37	84.10	73.12
	FDMatch	**89.23**	**93.53**	**79.54**	**85.88**	76.87	**85.01**	**74.42**
10% (97:879)	Onlysup	87.00	90.46	77.62	85.45	71.92	82.49	70.75
	Onlysup MTP	87.85	91.16	78.72	84.79	70.27	82.56	70.95
	FixMatch [33]	87.85	92.41	77.57	85.35	75.15	83.66	72.44
	UniMatch [34]	89.20	93.68	79.30	85.78	77.18	85.03	74.45
	CorrMatch [20]	88.67	92.76	78.88	85.65	73.70	83.93	72.91
	RankMatch [35]	89.26	93.65	79.20	85.56	77.69	85.07	74.50
	DWL [9]	89.13	93.90	79.15	85.72	76.97	84.97	74.39
	UCCL [21]	89.26	93.57	78.74	85.74	76.92	84.85	74.20
	FDMatch	**89.73**	**94.09**	**79.81**	**86.09**	**77.85**	**85.51**	**75.18**

TABLE IV

ABLATION STUDY OF THE PROPOSED FDMATCH FRAMEWORK WITH EXPERIMENTAL RESULTS OF 1% LABELED WHDLD AND LOVEDA

Method			WHDLD		LoveDA	
AFMD	ICRC	IPCC	MeanF1	mIoU	MeanF1	mIoU
			66.66	52.75	56.25	41.07
✓			68.98	55.28	61.64	45.15
	✓		68.47	54.68	58.62	43.02
		✓	67.66	54.09	58.77	43.30
✓	✓		69.27	55.60	62.68	46.44
✓		✓	69.35	55.75	62.51	46.77
	✓	✓	68.64	54.98	58.72	43.55
✓	✓	✓	**70.70**	**57.19**	**64.63**	**48.56**

TABLE V

ABLATION EXPERIMENTS OF N VALUE UNDER THE 5% LABEL RATIO OF THE VAIHINGEN DATASET

Representative features n	MeanF1	mIoU
n=16	84.66	73.93
n=32	84.74	74.05
n=64	**85.01**	**74.42**
n=128	84.91	74.30

2) Visualization Analysis of Each Module: To illustrate the impact of each component on feature representation learning, we conduct a visualization analysis using class activation mapping (CAM). Under the 1% labeled ratio of the LoveDA dataset, we compare the activation maps of baseline, baseline + AFMD, baseline + AFMD + ICRC, and the complete FDMatch framework, as shown in Fig. 6. The first two rows correspond to the building category, while the last two rows correspond to the forest category. For the building category, the introduction of AFMD significantly strengthens the activation responses within building regions, indicating that the foundation model distillation effectively enhances the model's class discrimination capability. The complete FDMatch framework further improves the detection of small-scale buildings, demonstrating a more refined spatial perception ability. For the forest category, the baseline exhibits weak activations under limited annotations. With the progressive inclusion of AFMD and ICRC, the activation intensity in forest regions increases markedly. Overall, the CAM visualizations clearly show that the incorporation of each module effectively enhances the model's feature response to different land-cover categories, thereby validating the effectiveness of the proposed method.

3) Effects of n in the IPCC Module: In the IPCC module, the number of representative features n serves as a key hyperparameter that influences model performance. When n is too small, the feature representation becomes insufficient and may lead to the loss of critical information. On the contrary, an excessively large n introduces redundant features and increases computational overhead. To identify an appropriate configuration, an ablation study is conducted on the Vaihingen dataset under the 5% labeled setting, as shown in Table V. The results indicate that the best performance is achieved when n = 64, which provides a favorable tradeoff between accuracy and computational efficiency. Therefore, n is set to 64 in all experiments.

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TABLE VI

ABLATION EXPERIMENTS OF FEATURE SAMPLING STRATEGY ON THE VAIHINGEN DATASET

Ratio	Sampling Strategy	mIoU	MeanF1
1%	Random	68.22	80.32
	IPCC	**69.01**	**81.11**
5%	Random	73.76	84.54
	IPCC	**74.42**	**85.01**
10%	Random	75.03	85.41
	IPCC	**75.18**	**85.51**

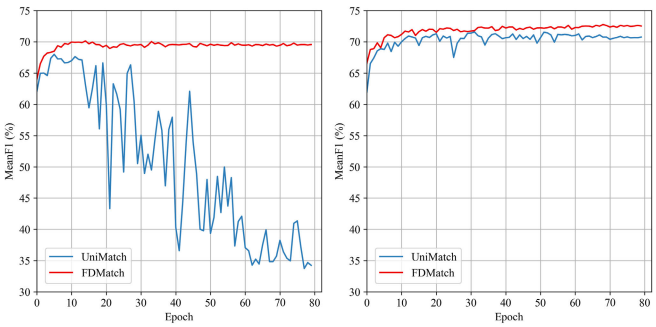


Fig. 7. MeanF1 of FDMatch and UniMatch on the Vaihingen validation set with 1% and 5% labeled data during training.

4) Effects of Feature Sampling Strategy in the IPCC Module: To evaluate the effectiveness of the representative feature sampling strategy in the IPCC module, a comparison is conducted with a random sampling strategy. In the random strategy, n features are randomly selected from the feature map to construct similarity matrices. The corresponding results are reported in Table VI. On the Vaihingen Dataset, under 1% and 5% labeled ratios, the proposed sampling strategy achieves improvements of 0.79% and 0.66% in terms of mIoU over random sampling, respectively. These results indicate that the proposed strategy effectively captures discriminative semantic information in the feature space, mitigates the semantic degradation associated with random sampling, and enhances the robustness of feature representations in scenarios with limited labeled data.

B. Training Convergence Analysis

In addition to performance evaluation, we further investigate the convergence behavior of the proposed model during training. On the Vaihingen dataset, we analyze the variation of MeanF1 accuracy on the validation set during training under 1% and 5% labeled ratios, comparing FDMatch with the baseline method UniMatch, as illustrated in Fig. 7. Under the extremely low 1% labeling setting, UniMatch reaches peak accuracy early but subsequently degrades due to accumulated pseudolabel noise. On the contrary, FDMatch leverages stable prior knowledge distilled from the RSFM to suppress noisy supervision, enabling continuous performance improvement and stronger robustness. When the labeled ratio increases to 5%, FDMatch achieves higher accuracy from the early stages and maintains its advantage throughout training with smaller performance fluctuations. Overall, compared with UniMatch, FDMatch achieves higher final accuracy, smoother convergence curves, and greater training stability, confirming its superior robustness in semisupervised remote sensing segmentation.

TABLE VII

RESULTS OF THE MODEL GENERALIZATION EXPERIMENTS ON THE WHDLD DATASET. MIOU IS REPORTED IN PERCENTAGE (%)

Ratio	Model	Method	mIoU
1%	Deeplabv3+	UniMatch	52.75
		FDMatch	**57.19**
	Unet	UniMatch	54.94
		FDMatch	**56.92**
	Segformer	UniMatch	54.00
		FDMatch	**57.32**
5%	Deeplabv3+	UniMatch	58.25
		FDMatch	**59.99**
	Unet	UniMatch	58.92
		FDMatch	**60.92**
	Segformer	UniMatch	59.31
		FDMatch	**60.94**
10%	Deeplabv3+	UniMatch	59.48
		FDMatch	**60.62**
	Unet	UniMatch	60.43
		FDMatch	**61.75**
	Segformer	UniMatch	60.30
		FDMatch	**61.34**

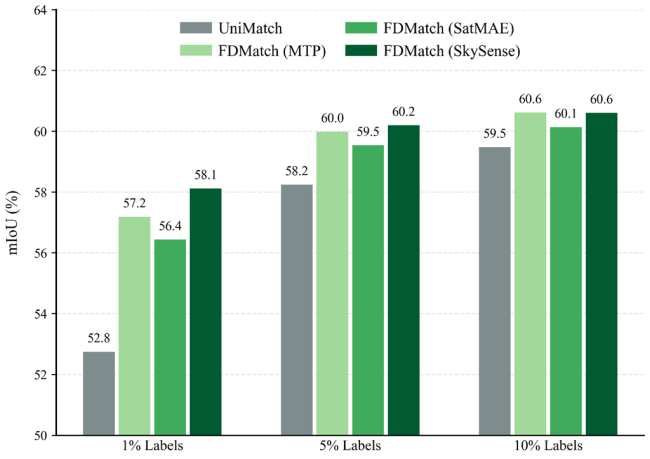


Fig. 8. Experimental results of the generalization capability of FDMatch across different foundation models on the WHDLD dataset.

C. Model Generalization Experiment

To evaluate the generalization capability of FDMatch across different segmentation architectures, we conducted experiments on the WHDLD dataset using DeepLabv3+ [28], UNet [27], and SegFormer [29]. Among them, DeepLabv3+ and UNet are representative CNN-based architectures with ResNet-50 as the encoder, while SegFormer is a Transformer-based architecture employing MiT-B2 as its encoder. We compared FDMatch with the baseline method UniMatch to assess the applicability and effectiveness of the proposed approach across diverse models. The experimental results are summarized in Table VII. It can be observed that FDMatch consistently outperforms UniMatch across all segmentation architectures. In particular, under the 1% labeled data setting, FDMatch achieves mIoU improvements of 4.44%,

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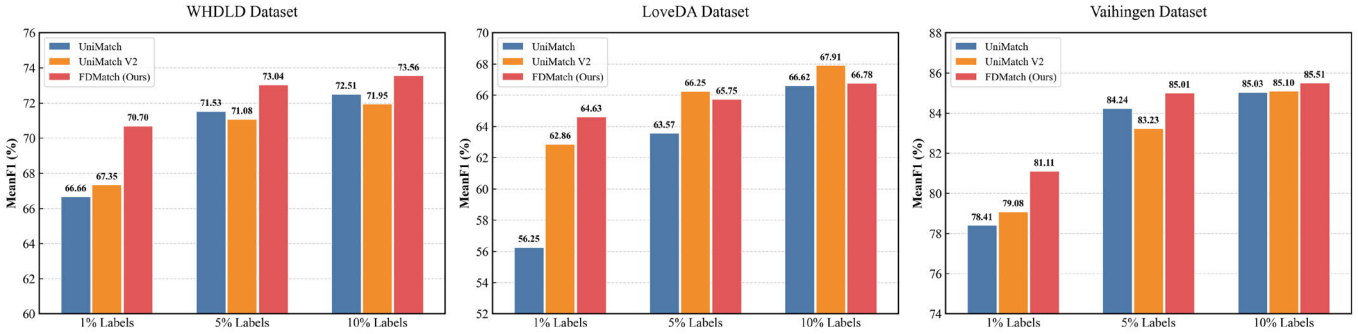


Fig. 9. MeanF1 of UniMatch, UniMatch V2, and FDMatch on different datasets with 1%, 5%, and 10% labeled data.

1.98%, and 3.32% on DeepLabv3+, UNet, and SegFormer, respectively, further demonstrating its robustness and generalizability across both CNN-based and Transformer-based frameworks.

Furthermore, to assess the generalization capability of FDMatch across different foundation models, MTP [46], SatMAE [45], and SkySense [51] are employed as knowledge sources, and experiments are conducted on the WHDLD dataset, with results presented in Fig. 8. The results indicate that FDMatch consistently outperforms UniMatch under all foundation model settings. Under the 1% labeled ratio, improvements of 4.4%, 3.6%, and 5.3% in terms of mIoU are achieved, which demonstrates strong adaptability to diverse foundation model architectures. Overall, FDMatch can be flexibly integrated with various foundation models and lightweight segmentation networks, yielding stable performance gains across different settings and exhibiting strong generalization capability.

D. Comparison With Foundation Model-Based Semisupervised Segmentation Methods

Foundation models trained on large-scale datasets have shown strong feature representation capabilities, and recent studies have begun to integrate them into semisupervised semantic segmentation. To clarify the differences between the proposed method and existing foundation model-based semisupervised approaches, we compare FDMatch with UniMatch V2 [52]. Different from conventional semisupervised frameworks that employ ResNet as the backbone network, UniMatch V2 adopts the pretrained foundation model DINOv2 as the encoder backbone to enhance feature representation capability. The corresponding experimental results are presented in Fig. 9.

On the relatively small WHDLD and Vaihingen datasets, UniMatch V2 achieves only limited performance improvement over UniMatch with a ResNet backbone and even shows performance degradation in some cases. This may be attributed to the large parameter scale of the foundation model backbone, which makes sufficient optimization difficult when labeled samples are scarce. On the contrary, FDMatch achieves stronger performance by transferring foundation model knowledge into a lightweight network and exploiting unlabeled data through multilevel consistency learning. On the larger LoveDA dataset, FDMatch still achieves the best performance under the 1% labeled ratio, whereas UniMatch V2 performs best under the 5% and 10% labeled ratios. This suggests that directly using a foundation model backbone can be effective when more labeled data are available, whereas FDMatch is more advantageous under extremely limited annotations. Overall, this comparison demonstrates that FDMatch differs from existing foundation model-based semisupervised methods not merely in the choice of backbone, but in the way foundation model knowledge is used. Rather than directly fine-tuning a large foundation model, FDMatch distills its remote sensing priors into a lightweight student network and combines this with multilevel consistency regularization. This design improves robustness under scarce labels while maintaining low inference overhead.

TABLE VIII
COMPARISON BETWEEN DIFFERENT METHODS IN TERMS OF COMPUTATIONAL COST

Method	Parameters	Memory	Training Speed	mIoU
UniMatch	**40.5 M**	**8.6 G**	**1.8 min/epoch**	52.75
CorrMatch	45.2 M	9.2 G	2.9 min/epoch	53.25
RankMatch	**40.5 M**	10.2 G	2.1 min/epoch	51.55
UCCL	**40.5 M**	14.6 G	3.1 min/epoch	53.50
FDMatch	**40.5 M**	13.2 G	3.1 min/epoch	**57.19**

E. Comparison of Computational Cost

A comparative analysis of computational cost is conducted between different semisupervised methods and FDMatch on the WHDLD dataset under the 1% labeled ratio, with results reported in Table VIII. Due to the incorporation of foundation model distillation, FDMatch results in higher GPU memory consumption during training. However, the training time remains comparable to that of CorrMatch and UCCL while achieving a notable performance gain with an mIoU improvement of 3.69%. It is important to note that the distillation process is applied only during training, with the parameters of the foundation model kept frozen. As a result, FDMatch maintains the same number of trainable parameters as other semisupervised methods and introduces no additional computational overhead during inference, thereby ensuring high inference efficiency.

VI. CONCLUSION

This article presents a semisupervised semantic segmentation framework for remote sensing images leveraging the knowledge distillation from RSFMs. The proposed method

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fully exploits the strong prior knowledge and transferability of foundation models by adaptively distilling their representations into a lightweight segmentation network, thereby improving accuracy under scarce annotations. To address the limitations of conventional pixel-level consistency regularization, the framework further introduces multilevel consistency constraints that enforce both ICRC and IPCC, enabling richer supervision from unlabeled data. Experiments on three benchmark datasets show that FDMatch consistently outperforms existing methods under various limited-label settings and exhibits strong generalization and training stability. Future work will explore more specialized distillation strategies tailored to RSFMs to reduce knowledge loss during transfer.

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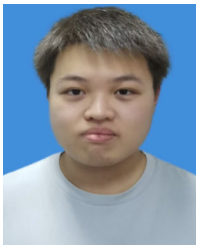
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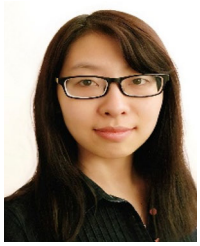
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